Physics-Informed Quantitative Forecasting in Finance

Project Specification

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1 Gap in Existing Provision

Forecasting in financial markets remains one of the most complex and uncertain problems in modern quantitative finance. Although advancements have been made through deep learning and time-series modelling, most trading algorithms still rely heavily on data-driven methods such as LSTMs or transformers. These models capture statistical correlations but struggle to adapt to shifts in market conditions, leading to a failure to generalise, as they often neglect the underlying structural dynamics that govern price movement, volatility and market momentum. The aim of this project is to bridge that gap by incorporating physics-informed neural networks (PINNs) to the realm of forecasting in financial markets. PINNs incorporate governing equations and domain-specific constraints directly into the loss function of neural networks, ensuring that model predictions respect known mathematical laws or physical analogues. By treating market evolution similarly to physical systems governed by stochastic differential equations (e.g. Black–Scholes, Ornstein–Uhlenbeck, Langevin dynamics), this project aims to enhance the interpretability and robustness of AI informed trading strategies. This project tackles the absence of theoretical grounding in AI models, by combining quantitative finance models with neural network architectures to yield data driven and theoretically sound forecasts.

2 Objectives

The primary objective of this project is to design, implement, and evaluate a physics informed deep learning framework which can forecast financial time-series data and underpin decisions in trading.

The system will first construct a dataset of historical financial indicators such as price, volume, volatility, and momentum factors. These indicators will be used to train a baseline deep learning model, likely an LSTM or Transformer architecture, aimed at modelling intricate time dependant relationships in market data.Building upon this baseline model, a physics informed regularisation term derived from quantitative finance equations (for example, stochastic differential equations or conservation-based constraints) will be integrated into the loss function to ensure that predictions are consistent with the theoretical financial dynamics.The trained model will then drive an AI trading agent that interprets forecasts to advise the decision of whether to buy, hold or sell. Its performance will be benchmarked against conventional machine-learning models and naïve trading strategies using various metrics such as mean squared error (MSE), Sharpe ratio, and directional accuracy.

If sufficient progress is achieved within this initial phase, the project will be extended to incorporate several optional features that enhance performance. These could include the integration of reinforcement learning to optimise sequential trading decisions under simulated environments, the expansion of the model to handle multi-asset forecasting and portfolio optimisation. It could further include an adoption of the system as a web application offering a user-friendly experience.Displaying real-time forecasts, confidence intervals and back-testing visualisations, users may indulge in interactively exploring predictions and assessing model performance. This extension would transform the project into an accessible and engaging AI based forecasting web application.

3 Methods and Methodology

The project will combine machine learning systems, quantitative modelling, and computational experimentation to develop a framework capable of providing physics informed financial forecasts. A hybrid Agile and Plan-Driven methodology will be adopted to ensure systematic yet responsive progress throughout development. The main components, including data processing, model implementation and evaluation, will follow a structured timeline. Other components, such as model tuning, experimentation and optimisation will be incrementally maintained through weekly sprints. To ensure consistency with academic objectives and sufficient progress, I will hold weekly meetings with my supervisor where we will review outcomes and adjust project priorities as needed.

Financial time series data, such as stock indices, currency pairs and market indicators, will be sourced via APIs like Yahoo Finance and Alpha Vantage. This will be followed by preprocessing steps including normalisation, stationarity testing and feature engineering to extract useful patterns in log returns, volatility and momentum factors. To efficiently retrieve and manage large volumes of time-stamped financial data, the system will employ a PostgreSQL database enhanced with the TimescaleDB extension for time-series optimisation. This will enable fast and structured queries for training, backtesting and feature generation. Supplementary analytics extracts will be stored in Parquet format for lightweight querying using DuckDB, while an optional Redis cache may be used for real-time price data within the web application.

The neural network architecture will be designed around temporal encoders such as LSTM, GRU or Transformer layers, with an additional physics-informed term embedded within the loss function to constrain predictions using theoretical finance equations. Model performance will be evaluated across both predictive and financial metrics (including RMSE, MAPE, Sharpe ratio, cumulative returns and maximum drawdown) with robustness validated through back-testing and Monte Carlo simulations.

Implementation will primarily use Python (TensorFlow/PyTorch, NumPy, Pandas) with Flask or Django powering a web application for visualising predictions, risk metrics and trading performance in real time. To ensure reproducibility and maintain professional software engineering standards, the project will employ GitHub for version control and integrate Docker for containerised development environments. A CI/CD pipeline will automate data ingestion, testing, model retraining and deployment, streamlining experimentation and preventing configuration drift. All experiments, methodologies, and results will be fully documented using LaTeX, ensuring an academically rigorous final dissertation.

4 Timetable

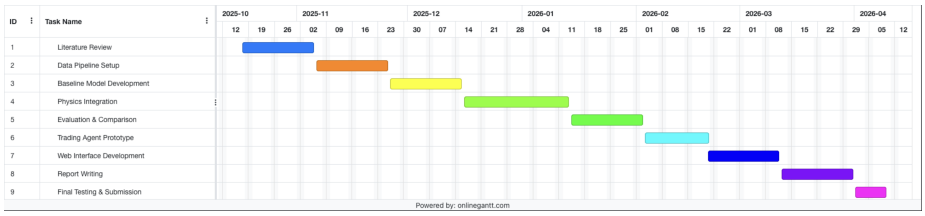

Figure 1: Project Gantt Chart

The project begins (from days 1–14) with a two-week literature review and planning phase. This aims to establish the theoretical foundations of PINNs and their application to financial modelling. During this stage, I will also set up the development environment, including Docker, GitHub version control and a CI pipeline. The result will be a clear research framework and scope agreed upon with my supervisor. If time is limited, I will narrow the focus to one core financial model and a smaller dataset.

The next phase, data pipeline development, will be from from days 15–28 and will involve acquiring, cleaning and storing market data using APIs such as Yahoo Finance or Alpha Vantage. Data will be stored in a PostgreSQL / TimescaleDB database for structured queries, with Parquet snapshots created for fast analysis. If setup delays occur, I will restrict data to a few key assets and rely on local storage instead of a full SQL implementation.

From days 29–42, I will implement a baseline deep-learning model to use as a benchmark for predictive performance. This stage will confirm the quality of the data pipeline and provide a reference point for later comparison. If training is delayed, I will focus solely on one LSTM model, rather than exploring various architectures. The core development period (from days 43–63) will focus on integrating the physics-informed component into the model and validating its impact on prediction accuracy. This is the critical research stage. Therefore, if progress slows, I will limit testing to one asset and timeframe to ensure a full set of evaluable results.

The evaluation and comparison phase (from days 64–77) will assess model performance using financial metrics such as cumulative returns and Sharpe ratio through systematic back-testing. If necessary, I can shorten this phase by reducing the number of assets or omitting secondary robustness tests.

Between days 78–91, I will build a prototype of the AI trading agent which uses the model's forecasts to simulate trading decisions. This will demonstrate how the framework could be transported practically in a real world trading scenario. In case of delays, I will keep this component rule-based, rather than implementing reinforcement learning.

The web application phase (from days 92–105) will involve deploying a dashboard using Flask or Django to visualise model predictions and trading results.

If there are time limitations, I will use a simpler Streamlit or static HTML version instead, or abandon it if absolutely necessary.

Finally, days 106–120 will be reserved for report writing, testing and submission. The complete system will be validated within Docker to ensure reproducibility before final packaging. Weekly meetings with my supervisor throughout the project will help identify risks early and adjust scope if required, ensuring that core components are prioritised and satisfactorily achieved.

5 Resources and Risk

This project relies on combining academic research materials, computational infrastructure, software tools and publicly accessible financial data sources. Each of these is essential to the successful development of the physics-informed forecasting system and suitable contingency measures have been planned in case of disruption.

Academic and Research Resources:
To identify appropriate quantitative equations and modelling techniques, I will make extensive use of academic databases such as IEEE Xplore, ScienceDirect, SpringerLink, arXiv, SSRN, and Google Scholar, alongside the University of Warwick Library's online journal repository. These platforms will be used to locate peer-reviewed papers in computational finance, stochastic processes, and physics-informed machine learning. The literature review will focus on established quantitative frameworks, including the Black–Scholes equation, Ornstein–Uhlenbeck process, and Langevin dynamics, as well as more recent studies applying physics-based constraints within neural networks. Google Scholar will be used as a discovery and citation-tracking tool to identify recent and highly cited works that may not yet appear in traditional databases. If any key papers are behind paywalls, I will access them through Warwick Library subscriptions, open-access preprints, or direct author contact. This approach ensures that all theoretical and mathematical choices are grounded in credible and current academic research.

Hardware and Software Resources:
Development and testing will primarily be conducted on a personal workstation with sufficient CPU and RAM capacity for deep-learning tasks. For computationally intensive training, I will use GPU acceleration via the University's high-performance computing cluster or Google Colab Pro. The project will be implemented using Python, with libraries such as TensorFlow or PyTorch for model training, Pandas and NumPy for data processing, and Matplotlib or Plotly for visualisation. The web interface will be developed in Flask or Django, and all dependencies will be managed within Docker containers to ensure reproducibility. GitHub will be used for version control and to run an integrated CI/CD pipeline for automated testing and deployment. In the event of hardware failure or GPU unavailability, model training will continue using reduced datasets or lighter architectures on CPU, maintaining the ability to meet project deadlines.

Data Resources:
Historical financial time-series data will be obtained through Yahoo Finance and Alpha Vantage APIs, which provide comprehensive open-source datasets on equities, indices, and currencies. Once collected, the data will be stored in a PostgreSQL database with the TimescaleDB extension for efficient time-series querying, with backup copies maintained in Parquet format for local offline access. If either API experiences downtime or access restrictions, I will use alternative sources such as Kaggle datasets or Quandl archives. Data caching and local backups will ensure that once data are downloaded, all experiments can be run offline without dependence on external providers.

Supervision and Development Infrastructure:
Weekly meetings with my project supervisor will be used to monitor progress, validate interim results, and adjust priorities where necessary. Development will take place within a Dockerised environment, ensuring consistency between local and cloud-based setups. This will prevent configuration issues and make it straightforward to reproduce or re-run experiments if technical problems occur.

Risks and Mitigation:
The main risks include computational delays, data unavailability, and time constraints.

- **Computational limitations:** Use smaller datasets or lower-frequency data; train models on CPU using reduced feature sets.
- **API or data access issues:** Rely on cached or open-source backups from Kaggle and Quandl.
- **Software incompatibility:** Switch between TensorFlow and PyTorch, or between Flask and Streamlit, as needed.
- **Time overruns:** Prioritise core deliverables (model and evaluation) over optional extensions such as reinforcement learning or advanced web deployment.
- **Model instability or overfitting:** Apply cross-validation, regularisation, and out-of-sample testing.

6 Legal, Social, Ethical and Professional Considerations

This project will rely solely on publicly accessible financial market data, excluding the use of confidential or human participant information. As such, ethical approval for the use of data is not required according to the University of Warwick's research ethics structure. However, the system generates financial forecasts and advises on trading therefore a disclaimer will be unequivocally displayed. This disclaimer will state that the web application is made for academic research purposes only, and should not be used as legitimate financial advise.

All data collection and storage will be in compliance with the University of Warwick's ethical guidelines, financial data usage policies, and relevant data licensing policies for APIs.The project will maintain transparency in data provenance, ensuring that every dataset is properly cited and used in accordance with its terms of service. Any external code or libraries incorporated will be open-source and appropriately accredited, following good software engineering and citation practices.

The project will also adhere to British Computer Society professional standards, promoting responsible use of AI in research. The work will be conducted with a strong emphasis on reproducibility, accuracy and intellectual honesty, ensuring that all reported findings and code are valid and can be independently verified. Additionally, all system outputs and documentation will respect principles of algorithmic transparency, explaining, where possible, how model predictions are generated and how uncertainty is represented, to uphold professional and ethical accountability in computational finance.